

HW

Trace quick sort as shown in the class

for the following array

85	76	99	13	45	103	-5	26	31
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350	719	400	200	100	96	900	890	760
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$i \mid j \mid a[i] \mid a[j] \mid a[\text{pivot}]$ $n =$
first =
last =
pivot =

(50)

a'

$n=9$ first = ~~0~~ last = ~~8~~ pivot = 0

swap
 $\text{85 76 99 13 45 103 -5 26 31}$
 $85 < 85$ $76 < 85$ $99 < 85$
 $\text{* swap } a[i], a[j]$
 $\text{85 76 31 13 45 103 -5 26 99}$
 $31 < 85$ $13 < 85$ $45 < 85$ $103 < 85$
 $\text{* swap } a[i], a[j]$

$\text{85 76 31 13 45 26 -5 103 99}$
 $26 < 85$ $31 < 85$ $103 < 85$
 $-5 > 85$ $103 > 85$

* Now swap $a[j]$ & $a[\text{pivot}]$

85 76 31 13 45 26 103 99
 $-5 < 85$ $76 < 85$ $31 < 85$ $13 < 85$ $45 < 85$ $26 < 85$
 $-5 > 85$ $76 > 85$
 $\text{* swap } a[j] \text{ \& } a[\text{pivot}]$
 99 103
 $103 < 99$ $99 < 103$
 $99 > 103$
 $\text{* swap } a[i] \text{ \& } a[\text{pivot}]$
 99 103
 $76 < 99$ $31 < 99$ $13 < 99$ $45 < 99$ $26 < 99$
 $26 > 99$

* swap $a[j]$ & $a[\text{pivot}]$

99 103 26 31 13 45 76
 $26 < 99$ $31 < 99$ $45 > 99$
 $13 > 99$

* swap $a[i]$ & $a[j]$

26 13 31 45
 $13 < 26$ $31 < 26$ $45 > 26$
 $13 > 26$

* swap $a[j]$ & $a[\text{pivot}]$

13 26 31 45

i	j	a[i]	a[j]	a(pivot)
0		85		85
1		76		85
2		99		85 X
	8		31	85 X
2		31		85
3		13		85
4		45		85
5		103		85 X
	8	99		85
	7	26		85 X

quicksort(a, 0, 5)

0	-5	-5	-5	-5
1	76	76		-5 X
	5		26	-5
	4		45	-5
	3		13	-5
	2		31	-5
	1		76	-5
	0		-5	-5 X

quicksort(a, 1, 5)

i	j	a[i]
1		76
2		31
3		13
4		45
5		26
	5	26

quicksort(a, 1, 4)

1		26
2		31
	4	45
	3	13
2		13
3		31
	3	31
	2	

a[i]

26

45

13

31

13

a[pivot]

~~76~~ 76

76

76

76

76

76 X

26

26 X

26

26 X

26 ✓

26 X

26 ✓

26 X

quicksort (a, 7, 8)

i	j	a[i]	a[j]	a(pivot)
				103
7		99		103 99
8		103		99 103
	8		103	99 103 X

at Finally

~~-5 26~~ ↑

Finally

-5 13 26 31 45 85 99 103

(1)

001 002 003

X

X

(2)

(3)

001 002 003

X

(4)

001 002 003

$\textcircled{i}$
 350 719 400 200 100 $\textcircled{j}$ 96 900 890 760
 ✓ ✗ ✓ ✓ ✓

$\textcircled{i}$ $\textcircled{j}$
 350 96 400 200 100 719 900 890 760
 ✗ ✗ ✗ ✓

$\textcircled{j}$ $\textcircled{i}$
 350 96 100 200 400 719 900 890 760
 ✓ ✓ ✗ ✓

swap $a[j]$ & $a[\text{pivot}]$

[200 96 100] 350 [400 719 900 890 760]

$\textcircled{i}$ $\textcircled{j}$
 200 96 100
 ✓ ✓ ✗
 ✗

swap $a[i]$ & $a[\text{pivot}]$

$\textcircled{i}$ $\textcircled{j}$
 [100 96] 200
 ✓ ✓
 ✗

swap $a[i]$ & $a[\text{pivot}]$

96 100 200

$\begin{matrix} \textcircled{j} & \textcircled{i} \\ 400 & 719 & 900 & 890 & 760 \\ \checkmark & \times & \checkmark & \checkmark & \checkmark \end{matrix}$

swap $a[i]$ & $a[\text{pivot}]$

400 719 900 890 760

$\begin{matrix} \textcircled{j} & \textcircled{i} \\ 719 & 900 & 890 & 760 \\ \checkmark & \times & \checkmark & \checkmark \\ \times & \checkmark & & \end{matrix}$

719 900 890 760

$\begin{matrix} \textcircled{i} & \textcircled{j} \\ 900 & 890 & 760 \\ \checkmark & \checkmark & \checkmark \\ & & \times \end{matrix}$

760 890 900

Finally.

96 100 200 350 400 719 760 890 900

i	j	a[i]	a[j]	a(pivot)
0		350		350
1		719		350 X
	8	7	760	350 ✓
	7		890	350 ✓
	6		900	350
	5		96	350 X

1		96		350
2		400		350 X
	5		719	350
			100	350 X
2		100		350 ✓
3		200		350
4		400		350 X
	5	400	400	350
	4	200	200	350 X

quicksort(a, 0, 2)

0	200	200 ✓
1	96	200

i	j	a[i]	a[j]	a(pivot)
2 100		100		200
	2	100	100	200 X

quicksort(a, 0, 4)

0		100		100 ✓
1		96		100 ✓
	1		96	100 X

quicksort(a, 4, 8)

4		400		400 ✓
5		719		400 X
	8	760	760	400
	7	890	890	400
	6	900	900	400
	5	719	719	400
	4	400	400	400 X

i	j	a[i]	a[j]	a[pivot]
quicksort(a, 5, 8)				
5		719		719
6		900		719 X
	8	760	769	719
	7	890	890	719
	6	900	900	719
	5	719	719	719 X

quicksort(a, 6, 8)

6		900		900
7		890		900
8		760		900 ✓
	8	760		900 X